

Manifold for Type 1 Diabetes

Pre-clinical risk stratification and intervention-window detection

T1D has a years-long pre-clinical prodrome. Current risk tools (TrialNet DPTRS, Index60, M120) give strong population survival curves but miss **individual-trajectory inflection points** — the window where a Stage-2-appropriate intervention has maximal benefit. **Manifold** is a causal-intelligence platform configured for T1D to detect those inflections: causal discovery on multi-omic cohorts, criticality estimators (DNB, Bayesian change-point, deep survival) feeding a composite fragility score (ΩF), and Pearl do-calculus / interdiction for principled therapy-selection framing.

Teplizumab's 2022 approval made the pre-clinical window *actionable*, not academic. Every month of better per-patient stratification is intervention-window bought.

ΩF PILLAR MAPPING - T1D PHYSIOLOGY

ΩF PILLAR	T1D READING
I — Interaction	Autoantibody-to-autoantibody coupling (IAA, GADA, IA-2A, ZnT8); HLA × viral-trigger epistasis
R — Reserve	β -cell functional mass proxied by C-peptide AUC on MMTT; glucose-response amplitude
J — Junction	Immune-metabolic subsystem coupling (transfer entropy between immune-cell subsets and glycemic variability)
C — Cascade	Autoreactive T-cell clonal expansion; epitope spreading rate
T — Temporal	Hazard of next stage transition given current trajectory

THE ASK

Partnership with a TEDDY-affiliated PI (or equivalent cohort steward — TrialNet, DiPiS, BabyDiab) for:

1. Retrospective data access under their existing Data Use Agreement.
2. Clinical co-authorship on the benchmark paper (ΩF vs. DPTRS, 10–12 weeks from data access).
3. Co-application to JDRF / Helmsley / NIH NIDDK for the v2 prospective arm.

Apex Analytica brings the causal-inference engine, the ΩF framework, and the software team. Full brief: M-T1D-01 .